SUNPOR — Status Brief

Architecture · Progress · Data · Prediction Timeline

1. How data enters the ML models

Live WinCC signals and operator forms feed one pipeline:

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WinCC (104 signals, Line E10)
→ Ingestion (~10s) → Cleaner → Rolling buffer → Feature engine (1/5/15/30 min)
→ Process State · Early Anomaly · Low-Production detail → Stored predictions
→ Operator forms + Daily Quality / Material Blocking → Learning over time
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Automatic data = machine behavior. **Operator forms** = human labels (events, observations, quality). Linked by production run.

2. Completed vs missing

Completed	Still missing
• Live ingestion + cleaning + features • Process State (Stable production confirmed on E10) • Early Anomaly Detection (Stages 0–2) • Low-Production cause & severity • Prediction storage + tests • Requirements Document (EN/DE)	• Client confirmation of other 8 phases • Consistent operator logging (knife, screen, nozzle, low production, quality) • Doser vibration / feed-in times (if available) • Enough labeled runs with later lab/quality outcomes • Material / Granulator / Nozzle risk models • Predictive Quality ML (needs delayed quality data)

3. Data collected so far — what we see

We collect: continuous Line E10 signals, features, and live predictions (process state, anomaly scores, low-production detail).

We see: stable production is detected reliably during normal runs; anomaly safety advisories can fire (e.g. process-water pressure); pipeline writes predictions live.

We do not yet have enough: operator-confirmed non-stable phases, maintenance events (knife/screen/nozzle), planned vs unplanned low production, and daily quality / blocking linked to runs. There was no historical dataset — learning starts from live capture.

4. When predictions become accurate

When	What becomes reliable	Needs
Now	Stable production; basic anomaly safety thresholds	Live signals (available)
~2–4 weeks*	Better Process State for other phases; useful low-production causes	Phase examples + Low Production events
~1–2 months*	Stronger early warnings; knife/screen/nozzle hints	Maintenance + material behavior labels
~2–3+ months*	First useful Predictive Quality during production	Daily quality + blocking linked to runs

*Depends on how consistently operators log events and how often phases / maintenance / quality outcomes occur.

5. Data we need — type and amount (minimum to start)

Data	Minimum	Better
Each process phase (confirmed examples)	5 per phase	15+
Low Production (with cause / planned–unplanned)	10	30+
Knife / blade change or grinding	10	20–30
Screen change	10	20–30
Nozzle change / flush / grind	10	20–30
Material behavior observations	15	40+
Daily quality (per shift / run)	20–30	ongoing
Material blocking	all cases	all cases

Example: ~10 knife/blade change entries is enough to start learning wear patterns; 20–30 makes it clearly more useful. Also needed: Material Production Status meaning, feeder mode meanings, and low-production / ΔP thresholds.

6. Bottom line

- Architecture is live: signals → features → Process State / Anomaly / Low-Production → predictions.
- Stable production works today; broader accuracy needs labeled operator and quality data.
- Ask: consistent logging — e.g. ~10 knife changes, ~10 screen changes, ~10 low-production events, phase examples, and daily quality each shift.